UDEMY AI/ML/DL/NLP COURSE

I bought this course but its certificate aint much valuable so just store its content which is available to be seen without purchase and study with claude

Udemy Link – [Complete Data Science,Machine Learning,DL,NLP Bootcamp]

GETTING STARTED

- VScode, Anaconda, Jupyter installations
- Python Basics
- Filing, Exceptions, Iterators, Generators.
- OOPs

ADVANCE PYTHON CONCEPTS

- Logging
- Multithreading and Multiprocessing
- Memory Management
- Flask and Streamlit

MATH

- Basic Statistics
- Basic Probability
- Inferential Statistics
- Probability Distributions
- Linear Algebra and Calculus (Overview)

DATA SCIENCE

EDA

- Pandas
- Numpy
- Matplotlib
- Seaborn
- Basic EDA

FEATURE ENGINEERING

- Handling Outliers, Missing and Imbalanced Data
- SMOTE
- Data Encoding (Normal, Label, Ordinal, Target Guided)

PRACTICE

- Red Wine Dataset
- Flight Price Dataset
- Google Play Store Dataset

MACHINE LEARNING

BASICS

- Model and Instance based approach
- Training and Testing (split)
- Overfitting and Underfitting, Bias/Variance Tradeoff

- Parameter Estimation (MLE, Laplace, Linear Interpolation, Tuning, Errors/Baselines, Confidence)
- Cross Validation (LOOCV, KFOLD, Time Series, Random State)
- Okham's Razor Principle
- Hyper Parameter Tuning
- Evaluation Metrics (Accuracy, Precision, Recall, F1, Confusion Matrix, F-beta score)

SUPERVISED – REGRESSION

- Linear Regression (Equation, Multiple LR, Polynomial LR, Cost Function, Convergence Algo, Mean Squared Error, OLS, Implementation and Pipeline)
- Ridge, Lasso, Elasticnet Regression (Its the hyperparameter tuning of LR)
- Model Training Project (Data Preprocessing → Feature Engineering → EDA → Basic LR → Evaluation Metrics → CV and Hyperparameter Tuning → Deployment via Flask/AWS)
- Logistic Regression (Equation, Performance metrics, OVR, ROC curve, Imbalanced data, Implementation)
- Grid Search and Randomised Search (Its hyperparameter tuning of logistic regression)

SUPERVISED – CLASSIFICATION

- Support Vector Machines (Soft Margin & Hard Margin, Maths Intuition, SVC Cost Function, SVM Kernels, Support Vector Classifiers, Support Vector Regression, Implementation)
- Naive Bayes Theorem (Bayes' Theorem, Variants of Naive Bayes, Practical Implementation)
- K Nearest Neighbour (Classification & Regression Indepth Intuition, Optimization using KD Tree and Ball Tree, Classifier and Regressor Implementation)
- Decision Tree Classifier and Regressor (Entropy and Gini Impurity, Information Gain, Splits for Numerical Features, Pre-Pruning & Post-Pruning, Regression, Implementation)
- Random Forest (Bagging & Boosting Ensemble Techniques, Regression, Problem Classification, Feature Engineering, Model Training)
- Adaboost (Decision Tree Stump Creation & Performance, Updating and Normalising Weights, Selecting Datapoints for Next Tree, Final Prediction, Model Training)
- Gradient Boosting (Regression, Classifier Training, Regression Model Training)
- Xgboost (Classification Indepth Intuition, Regressor, Model Training)

UNSUPERVISED

- Introduction to Unsupervised Machine Learning
- PCA (Curse of Dimensionality, Feature Selection and Extraction, Geometric Intuition, Maths Intuition, Eigen Decomposition on Covariance Matrix, Implementation)
- K Means Clustering (Geometric Intuition, Finding K Values, Random Initialisation Trap/K-means++, Implementation)
- Hierarchical Clustering (Agglomerative Clustering Implementation, K-means vs Hierarchical Clustering)
- DBSCAN Clustering (How DBSCAN Works, Pros and Cons, Implementation)
- Silhouette Clustering (Intuition)

DEPLOYMENT & VERSION CONTROL

DOCKER & GIT

- Dockers For Beginners (What is Containers, Docker Image vs Containers, Docker vs Virtual Machines, Installation, Basic Commands, Creating & Pushing a Docker Image, Docker Compose)
- Git For Beginners (Introduction, Installation and Basic Commands, Merge/Push/Checkout/Log Commands, Resolving Branch Merge Conflicts)

END TO END PROJECT

End to End ML Project With AWS, Azure Deployment (Github Repository & Code Setup, Project Structure/Logging/Exception Handling, Problem Statement/EDA/Model Training, Data Ingestion, Data

Transformation Pipelines, Model Trainer & Hyperparameter Tuning, Prediction Pipeline, Deployment via AWS Beanstalk, EC2 with ECR, and Azure Container Deployment)

MLOPS

- End to End MLOPS Project With ETL Pipelines – Network Security System (Project Structure & Environment Setup, Github Repo Setup, Packaging with Setup.py, Logging & Exception Handling, ETL Pipeline Introduction & Setup, Data Ingestion Architecture & Implementation, Data Validation, Data Transformation, Model Trainer & Evaluation with Hyperparameter Tuning, MLFlow Experiment Tracking with Dagshub, Model Pusher, Training & Batch Prediction Pipelines, Artifact Push to AWS S3, Docker Image & Github Actions, ECR Push, Final EC2 Deployment)
- MLFlow, Dagshub and BentoML – Complete ML Project Lifecycle (Getting Started with MLOps using MLFlow & Dagshub, Data Versioning Control, Building Data Science Projects with BentoML)

NLP FOR MACHINE LEARNING

- Roadmap and Practical Use Cases of NLP
- Tokenisation and Text Preprocessing (Basic Terminologies, Stemming, Lemmatization, Stopwords, Parts of Speech Tagging, Named Entity Recognition)
- Text Representation Techniques (One Hot Encoding, Bag of Words, N-Grams, TF-IDF – Intuition, Advantages/Disadvantages, Implementation using NLTK)
- Word Embeddings (Word2Vec Intuition, CBOW, SkipGram, AvgWord2Vec, Practical Implementation with Gensim)
- Practical Projects (Spam Ham Classification using BOW & TF-IDF, Text Classification with Word2Vec & AvgWord2Vec, Kindle Review Sentiment Analysis)
- Role Play Practice Scenarios (NLP Engineer at a News Aggregation Startup, Data Scientist at a Movie Review Platform)

DEEP LEARNING

ANN FUNDAMENTALS

- Perceptron and ANN (Why Deep Learning is Popular, Perceptron Intuition & Pros/Cons, ANN Intuition and Learning, Back Propagation & Weight Updation, Chain Rule of Derivatives, Vanishing Gradient Problem)
- Activation Functions (Sigmoid, Tanh, Relu, Leaky Relu, Parametric Relu, ELU, Softmax for Multiclass Classification, Choosing the Right Activation Function)
- Loss and Cost Functions (Loss vs Cost Function, Regression Cost Function, Classification Loss Function, Choosing the Right Loss Function)
- Gradient Descent Optimisers (SGD, Mini Batch SGD, SGD with Momentum, Adagrad, RMSProp, Adam Optimiser, Exploding Gradient Problem, Weight Initialisation Techniques, Dropout Layers)

CNN

Convolutional Neural Networks (Human Brain vs CNN, Image Fundamentals, Convolution Operation, Padding, CNN vs ANN, Max/Min/Average Pooling, Flattening & Fully Connected Layers, CNN Example with RGB)

END TO END DEEP LEARNING PROJECT (ANN)

End to End Deep Learning Project Using ANN (Classification Problem Statement, Feature Transformation with Sklearn, Training with Optimizer & Loss Functions, Prediction, Streamlit Web App Integration & Deployment, ANN Regression Implementation, Finding Optimal Hidden Layers & Neurons)

RNN AND VARIANTS

- NLP with Deep Learning (Introduction)
- Simple RNN (Architecture – RNN vs ANN, Forward & Backward Propagation Through Time, Problems with RNN)

- End to End Deep Learning Project With Simple RNN (Problem Statement, Word Embedding Layers, Implementation with Keras Tensorflow, IMDB Dataset & Feature Engineering, Training, Prediction, Streamlit Deployment)
- LSTM and GRU RNN (Why LSTM, Architecture, Forget/Input/Output Gates, Training Process, Variants, GRU Complete Intuition)
- LSTM and GRU End to End Project – Predicting Next Word (Problem Statement, Data Collection & Processing, Model Training, Prediction, Streamlit Integration, GRU Variant Implementation)
- Bidirectional RNN Architecture and Indepth Intuition

SEQUENCE MODELS & TRANSFORMERS

- Encoder Decoder | Sequence to Sequence Architecture (Indepth Intuition, Problems with Encoder-Decoder)
- Attention Mechanism – Seq2Seq Networks (Indepth Architecture Explanation)
- Transformers (Plan of Action, Why Transformers, Basic Architecture, Self Attention Layer, Multi Head Attention, Feed Forward Network, Positional Encoding, Layer Normalization, Complete Encoder Architecture, Decoder – Masked Multi Head Attention, Encoder-Decoder Multi Head Attention, Final Linear & Softmax Layer)

CLAUDE CODE

Claude Code – Vibe Coding for Developers (Claude Ecosystem, Installation and Working, Building Agents, Agent Views & Teams, Hooks, Skills and Plugins)

Priority	Block	Why
Now	Finish Classification (SVM → XGBoost) + Unsupervised	Continues your current momentum, cheap on RAM
Now	NLP basics + 1–2 projects	Cheap, high interview-signal
Defer	Full DL (CNN/RNN/Transformers)	Heavy compute, long payoff, do post-internship or in a dedicated block
Defer/skim only	MLOps (MLFlow/Dagshub/BentoML)	Not what a startup screen tests for at your stage